

TGDetect - A Temporal Graph Neural Network Framework for AI-Based Multi-Step Cyberattack Detection

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PROBLEM DEFINITION

Current security systems analyze log events independently, making it difficult to detect cyberattacks that unfold through multiple time-separated actions. They often fail to preserve relationships between users, devices, files, and processes, ignore event timing, and generate fragmented alerts. As a result, advanced attacks such as APTs, Living-off-the-Land, and zero-day threats may go undetected. TGDetect overcomes these limitations by modeling security events as an evolving temporal graph and using a Temporal Graph Neural Network (TGNN) to correlate related events, detect multi-stage attacks, explain suspicious behavior, and reconstruct the complete attack timeline for faster incident response.

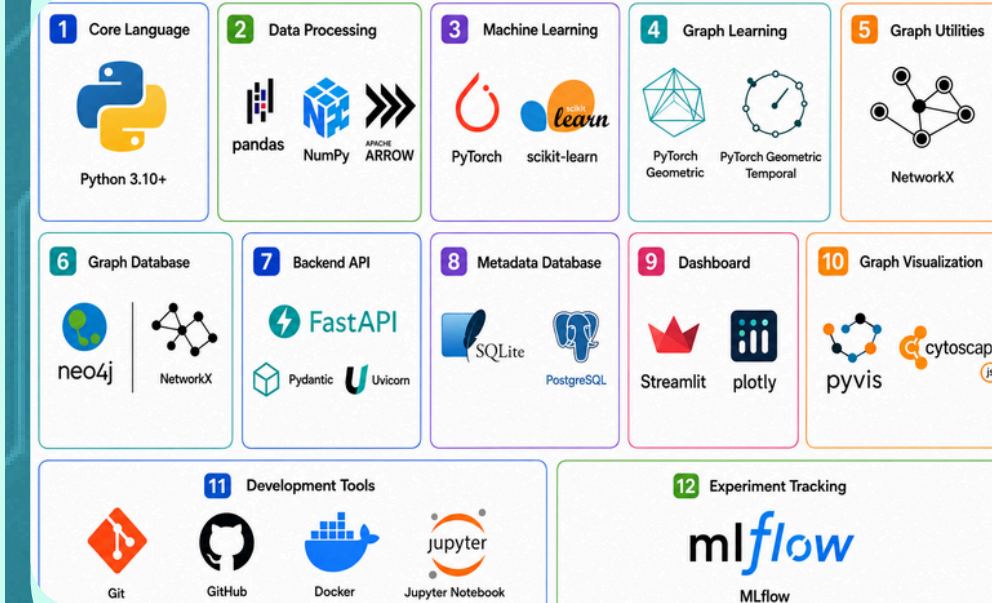
OBJECTIVES

- To develop a heterogeneous continuous-time Temporal Graph Neural Network (TGNN) for modeling security events and their temporal relationships.
- To incorporate online concept drift adaptation for continuously updating the TGNN model in response to evolving attack behaviors.
- To design an attack backtracking engine that automatically reconstructs the complete attack chain from the generated alert to the initial point of compromise.
- To provide temporal explainability by mapping detected attack patterns and model attention to the MITRE ATT&CK framework.
- To integrate threat intelligence feeds with graph-based analysis to improve detection accuracy and contextual understanding of cyber threats.

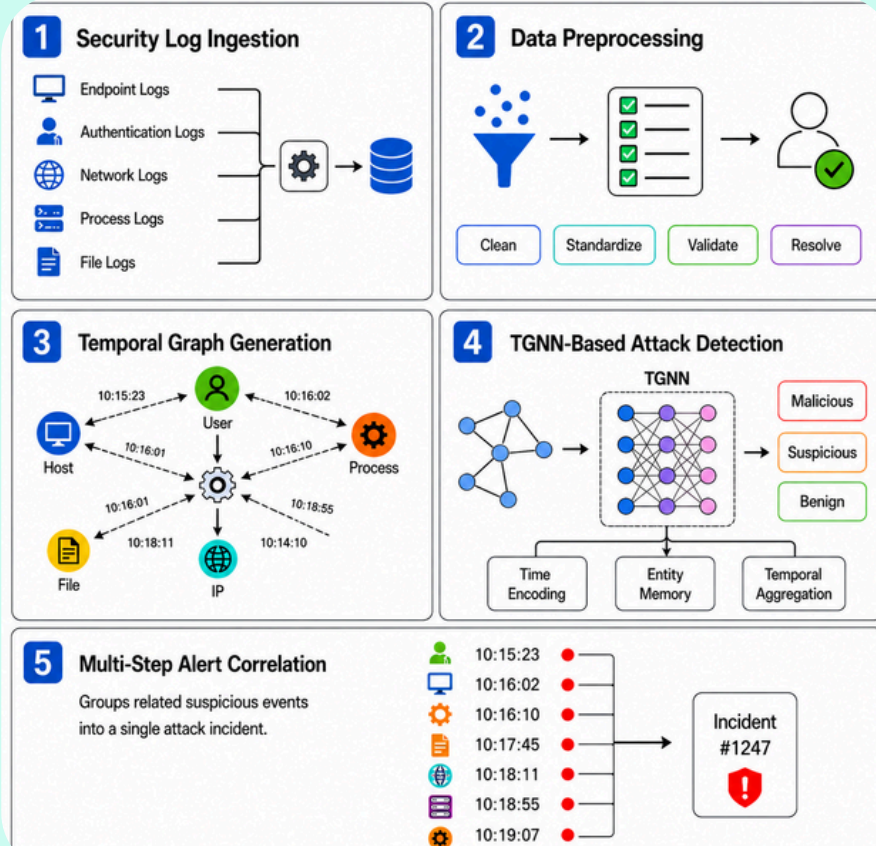
SCOPE

- Can be used to collect and preprocess security logs from multiple sources into a unified format.
- Can be used to build an evolving temporal graph with users, hosts, files, processes, and network entities.
- Can be used to implement a Temporal Graph Neural Network (TGNN) for analyzing event relationships over time.
- Can be used to classify events and entities as benign, suspicious, or malicious and generate dynamic risk scores.
- Can be used to correlate related events into a single security incident and reconstruct the attack path.

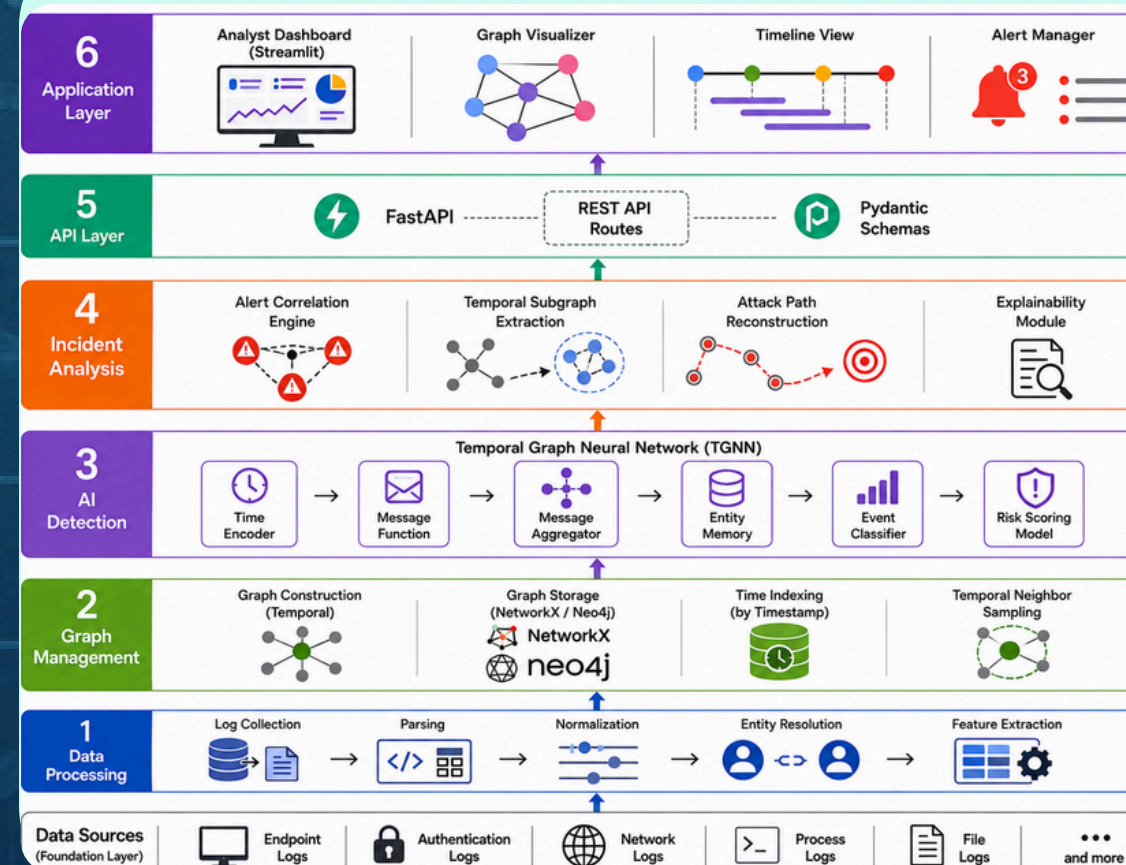
Technology stack



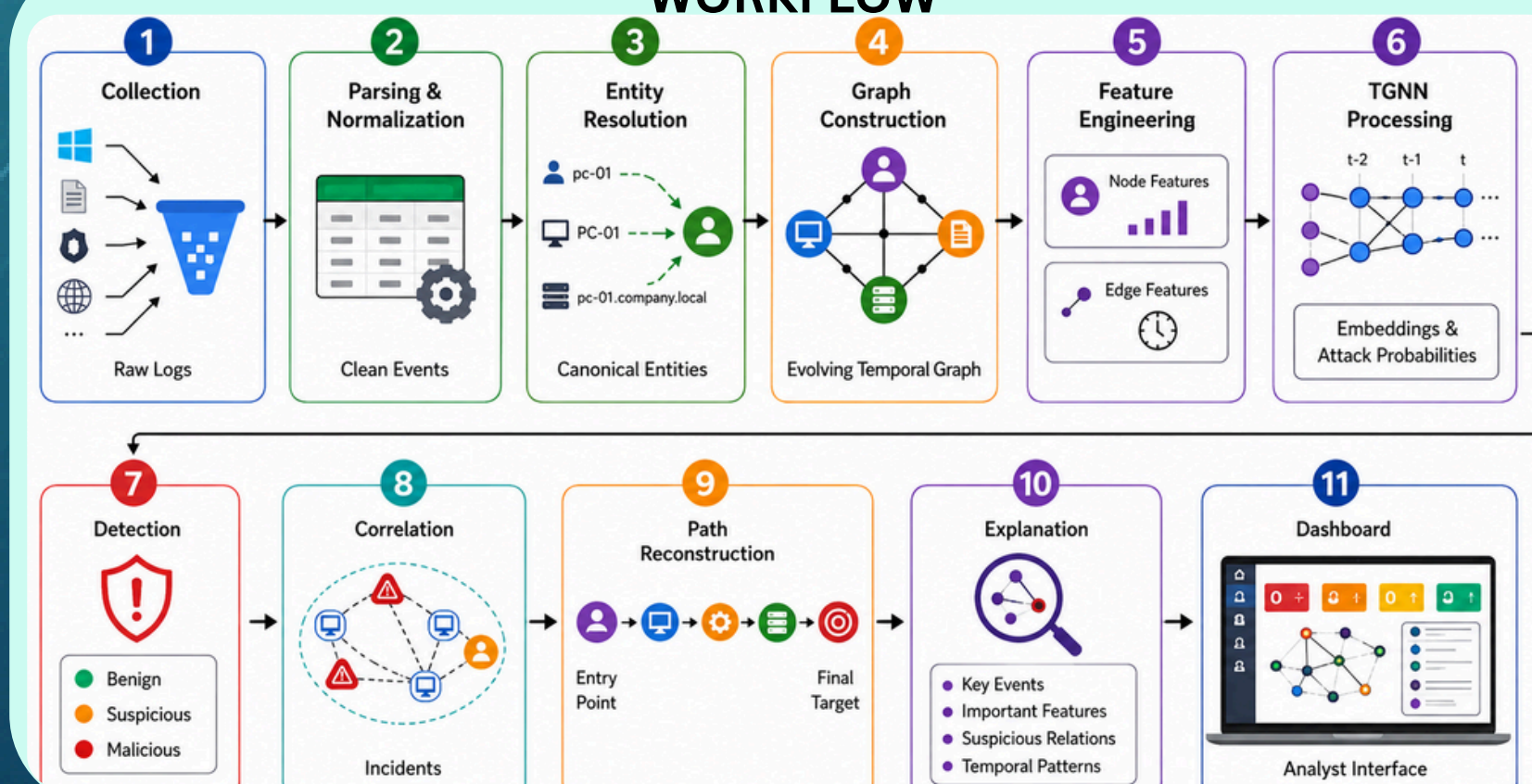
FEATURES



SYSTEM ARCHITECTURE



WORKFLOW



References: 1)TAPAS: An Efficient Online APT Detection withTask-guided Process Provenance GraphSegmentation and Analysis, 2) MAGIC: Detecting Advanced Persistent Threats via Masked Graph Representation Learning, 3) Huntrace: Graph Neural Network Based APT Intrusion Detection on Homogeneous Provenance Graphs.